November 2026 sky events and Earth activity: a test against past clusters

Research brief Research date: 6 November 2026 · Earthquake data to 6 Nov 2026 · Eruption data to 31 Dec 2025 · Method and code: tools/sky/ in this site's repository

> **Status.** Historical comparison complete. The November 2026 window is prospective; results will be added after 1 December 2026.

1. Bottom line

No unusual earthquake or eruption activity was found around past sky clusters like November 2026. Six periods in 1973–2026 meet the rule fixed in advance: at least four planets grouped in the sky, a supermoon and a Mars–Jupiter conjunction, all within 30 days. Five are in the past: April 2011, October 2015, January 2018, March 2020 and May–June 2022. In those windows, global M6+ earthquake counts, M7+ counts and eruption onsets all stayed within the normal range for windows of the same length. None of the 16 primary tests survives correction for multiple testing (lowest q = 0.77). The same holds for each feature on its own: 27 Mars–Jupiter conjunctions, 146 supermoons and 67 planet-grouping episodes.

One modest anomaly appears: five eruption onsets in the May–June 2022 window, against a typical 1.9 (97th percentile). With 24 period-by-outcome comparisons, about one such result is expected by chance. The November 2026 window (13–27 November) has not happened yet. Results to date (1–6 November) are observations only, not a test.

Four separate statements apply. An event happening is not the same as it being statistically unusual. Being unusual is not the same as being close in time to an astronomical event, and being close in time is not evidence that the astronomy contributed. This brief makes no earthquake or eruption forecast [9].

2. Verified astronomical timeline (UTC)

Every date was computed independently, not taken from headlines. Planet and Moon positions come from PyEphem. The comet comes from the current JPL orbit and the meteor shower from the IMO calendar. Two corrections to circulating dates follow. The Mars–Jupiter conjunction is on 16 November UTC (15 November in the Americas), not the 14th. Comet 161P's perihelion is 27 November per JPL's October 2026 orbit; some websites still list an older 18 December date. An apparent alignment or conjunction is a line-of-sight effect: Mars and Jupiter are about 4 AU apart in space on 16 November.

Date (UTC)	Event	Definition / note	Source
4 Nov 14:24	Mercury inferior conjunction (0.67 AU)	Mercury between Earth and Sun	PyEphem [10]
10 Nov – 4 Dec (interval)	Morning grouping: Mercury, Venus, Mars, Jupiter	≥4 naked-eye planets west of the Sun, each ≥10° from it, within 90° of longitude. Best seen 13–22 Nov	PyEphem [10]
13 Nov 15:55 / 14 Nov 00:32	Mercury, then Venus, turn direct	Apparent motion only	PyEphem [10]
16 Nov 06:24	Mars–Jupiter conjunction, 1.2° apart, 25.5° Leo	Equal ecliptic longitude seen from Earth. The planets are about 4 AU apart in space	PyEphem [10]
17 Nov 23:45 (nodal peak)	Leonids, ZHR about 15; weak dust-trail encounters 17–19 Nov	Meteors from comet 55P/Tempel–Tuttle debris	IMO [2]
24 Nov 14:53	Full Moon at 360,765 km: a supermoon	Full Moon ≤ 362,000 km (this study). Perigee 25 Nov 20:54 UTC at 359,347 km	PyEphem [10]
25 Nov 22:24	Uranus at opposition	Earth between Sun and Uranus	PyEphem [10]
27 Nov ≈18:17 TDB	Comet 161P/Hartley–IRAS at perihelion, q = 1.28 AU	Closest to Earth was about 2 Oct 2026 (≈0.52 AU, two-body estimate from the JPL orbit). Predicted magnitude about 17 in late November: a telescope object	JPL SBDB [1]
30 Nov	Moon passes Jupiter (08:34) and Mars (17:24) near Regulus	Apparent grouping	PyEphem [10]

3. Design, fixed before the data were examined

- **Grouping:** at least 4 of Mercury, Venus, Mars, Jupiter and Saturn on the same side of the Sun, each at least 10° from it, within 90° of ecliptic longitude (daily, 00 UTC). **Supermoon:** full Moon at most 362,000 km away (about three a year). **Conjunction:** equal geocentric longitude of Mars and Jupiter. **Cluster:** all three within one 30-day span. These definitions were fixed before the catalogues were opened.
- **Earthquakes:** USGS ComCat global M6.0+, 1973 to 6 November 2026 [3]. Counts are reported for all events and for independent mainshocks; aftershocks were removed with Gardner–Knopoff space–time windows [5]. M6+ is complete globally for this period. Regional swarms were not assessed, because they need local catalogues and local baselines.
- **Volcanoes:** Smithsonian GVP confirmed eruption onsets dated to within ±3 days, 1973–2025 [4]; 2026 is left out because the database lags. The record has no

consistent escalation or unrest entries, so those are assessed as *insufficient evidence*, not as no effect.

- **Windows:** ±3 days around each event (primary), with ±1 and ±7 days as sensitivity checks. A grouping counts as its whole run, padded by 3 days. Overlapping windows are merged, so no earthquake is counted twice.
- **Baselines:** (a) 5,000 circular time shifts of the event dates, which keep aftershock and seasonal clustering, giving a one-sided p and Benjamini–Hochberg q across the 16 primary tests; (b) every same-length window in 1973–2025; (c) the same calendar window in all other years.

4. Findings by period

All historical periods that meet the rule are listed, plus the August 2024 candidate from the original brief. August 2024 fails the grouping rule: its "six-planet alignment" was spread over more than 90° and included Uranus and Neptune, which are not visible to the naked eye. The rule also finds three clusters the brief did not list (2011, 2018, 2020). Percentile = where the observed count sits among all same-length windows in 1973–2025.

Astronomical event / window	Terrestrial observation	Expected baseline (mean, 95% range)	Degree of anomaly	Data quality	Interpretation
Apr 2011 2011-04-15 to 2011-05-04; Mars–Jupiter 01 May; supermoon 18 Apr (358,796 km); morning grouping 21 Apr–04 Jun (Mercury, Venus, Mars, Jupiter)	5 M6+ (3 mainshocks, 0 M7+); 2 eruption onsets. Largest: M6.8 79 km W of Kirakira, Solomon Islands (23 Apr)	5.5 mainshocks (95%: 1–12); 0.7 M7+; 1.6 onsets (95%: 0–5)	mainshocks 18th, M7+ 26th, eruptions 65th percentile	Good (global M6+ complete since 1973; GVP onsets ±3 d)	Within normal range
Oct 2015 2015-10-14 to 2015-10-30; Mars–Jupiter 17 Oct; supermoon 27 Oct (359,328 km); morning grouping 06 Oct–01 Nov (Mercury, Venus, Mars, Jupiter)	6 M6+ (6 mainshocks, 2 M7+); 3 eruption onsets. Largest: M7.5 Hindu Kush region, Afghanistan (26 Oct)	4.7 mainshocks (95%: 1–10); 0.6 M7+; 1.4 onsets (95%: 0–4)	mainshocks 72th, M7+ 93th, eruptions 89th percentile	Good (global M6+ complete since 1973; GVP onsets ±3 d)	Within normal range

Astronomical event / window	Terrestrial observation	Expected baseline (mean, 95% range)	Degree of anomaly	Data quality	Interpretation
Jan 2018 2017-12-30 to 2018-01-10; Mars–Jupiter 07 Jan; supermoon 02 Jan (356,602 km); morning grouping 02 Jan–03 Feb (Mercury, Mars, Jupiter, Saturn)	1 M6+ (1 mainshocks, 1 M7+); 1 eruption onsets. Largest: M7.5 203 km NNE of Barra Patuca, Honduras (10 Jan)	3.3 mainshocks (95%: 0–8); 0.4 M7+; 1.0 onsets (95%: 0–3)	mainshocks 12th, M7+ 80th, eruptions 56th percentile	Good (global M6+ complete since 1973; GVP onsets ±3 d)	Within normal range
Mar 2020 2020-03-06 to 2020-03-23; Mars–Jupiter 20 Mar; supermoon 09 Mar (357,400 km); morning grouping 03 Mar–25 Apr (Mercury, Mars, Jupiter, Saturn)	6 M6+ (5 mainshocks, 0 M7+); 1 eruption onsets. Largest: M6.4 Kermadec Islands region (14 Mar)	5.0 mainshocks (95%: 1–11); 0.6 M7+; 1.5 onsets (95%: 0–4)	mainshocks 54th, M7+ 28th, eruptions 40th percentile	Good (global M6+ complete since 1973; GVP onsets ±3 d)	Within normal range
May 2022 2022-05-26 to 2022-06-17; Mars–Jupiter 29 May; supermoon 14 Jun (357,656 km); morning grouping 17 Feb–08 Jul (Mercury, Venus, Mars, Jupiter, Saturn)	8 M6+ (7 mainshocks, 1 M7+); 5 eruption onsets. Largest: M7.2 10 km W of Azángaro, Peru (26 May)	6.4 mainshocks (95%: 1–13); 0.8 M7+; 1.9 onsets (95%: 0–5)	mainshocks 61th, M7+ 64th, eruptions 97th percentile	Good (global M6+ complete since 1973; GVP onsets ±3 d)	Eruption onsets in the top 5% of same-length windows: a modest anomaly, one of 24 comparisons
Nov 2026 2026-11-13 to 2026-11-27; Mars–Jupiter 16 Nov; supermoon 24 Nov (360,765 km); morning grouping 10 Nov–04 Dec (Mercury, Venus, Mars, Jupiter)	Window 13–27 Nov not yet observed	4.2 mainshocks (95%: 0–9); 0.5 M7+; 1.2 onsets (95%: 0–4)	—	Good (global M6+ complete since 1973; GVP onsets ±3 d)	Prospective; see monitoring plan

Astronomical event / window	Terrestrial observation	Expected baseline (mean, 95% range)	Degree of anomaly	Data quality	Interpretation
Aug 2024 (brief's candidate) 2024-08-11 to 2024-08-31; Mars–Jupiter 14 Aug; full Moon 19 Aug 361,969 km (no qualifying grouping)	6 M6+ (3 mainshocks, 1 M7+); 0 eruption onsets. Largest: M7.0 102 km E of Petropavlovsk-Kamchatsky, Russia (17 Aug)	5.8 mainshocks (95%: 1–12); 0.7 M7+; 1.7 onsets (95%: 0–5)	mainshocks 16th, M7+ 67th, eruptions 9th percentile	Good (global M6+ complete since 1973; GVP onsets ±3 d)	Within normal range

5. Each feature tested on its own, 1973–2026

Keeping the three features separate matters because they are physically different: a tidal maximum and two line-of-sight effects. Ratios of observed to expected counts sit between 0.67 and 1.65. None is significant after correction.

Feature (±3 d)	Outcome	Observed	Expected	Ratio	p	q	Verdict
Mars–Jupiter conjunction	M6+ earthquakes, all	70	68.7	1.02	0.444	0.77	No excess
Mars–Jupiter conjunction	M6+ mainshocks	55	50.6	1.09	0.293	0.77	No excess
Mars–Jupiter conjunction	M7+ mainshocks	7	6.3	1.11	0.45	0.77	No excess
Mars–Jupiter conjunction	Eruption onsets	10	14.8	0.67	0.922	0.92	No excess
Supermoon (full Moon ≤ 362,000 km)	M6+ earthquakes, all	362	382.2	0.95	0.782	0.89	No excess
Supermoon (full Moon ≤ 362,000 km)	M6+ mainshocks	274	279.5	0.98	0.615	0.82	No excess
Supermoon (full Moon ≤ 362,000 km)	M7+ mainshocks	39	34.7	1.12	0.24	0.77	No excess
Supermoon (full Moon ≤ 362,000 km)	Eruption onsets	82	82.2	1.0	0.52	0.77	No excess
Planet grouping (≥4 planets, whole run)	M6+ earthquakes, all	948	953.6	0.99	0.53	0.77	No excess
Planet grouping (≥4 planets, whole run)	M6+ mainshocks	703	696.9	1.01	0.427	0.77	No excess

Feature (±3 d)	Outcome	Observed	Expected	Ratio	p	q	Verdict
Planet grouping (≥4 planets, whole run)	M7+ mainshocks	93	86.5	1.08	0.242	0.77	No excess
Planet grouping (≥4 planets, whole run)	Eruption onsets	220	205.2	1.07	0.153	0.77	No excess
All three within 30 days (cluster span)	M6+ earthquakes, all	26	34.2	0.76	0.84	0.9	No excess
All three within 30 days (cluster span)	M6+ mainshocks	22	24.8	0.89	0.687	0.85	No excess
All three within 30 days (cluster span)	M7+ mainshocks	4	3.0	1.32	0.367	0.77	No excess
All three within 30 days (cluster span)	Eruption onsets	12	7.3	1.65	0.079	0.77	No excess

Full results, including ±1 and ±7 day sensitivity checks: sky_cluster_tests.csv; periods: sky_cluster_periods.csv.

6. Strongest and negative findings

- **Modest anomaly.** May–June 2022 had five eruption onsets (Dempo, Ulawun, Bulusan, Mayon, Ebeko) where 1.9 is typical. These are ordinary small eruptions (VEI 0–2) in arcs that erupt often. Across all six cluster windows, onsets were 12 against 7.3 expected (p = 0.08, q = 0.77): suggestive, not significant.
- **October 2015** had two M7+ mainshocks in 17 days (93rd percentile), including the M7.5 Hindu Kush earthquake of 26 October.
- **Exploratory only.** Sensitivity checks with p < 0.1: Mars–Jupiter conjunction ±1 d, M6+ mainshocks: 32 vs 21.7 (p = 0.035); All three within 30 days (cluster span) ±1 d, Eruption onsets: 10 vs 5.7 (p = 0.07). These were not pre-specified as primary, do not hold at the primary ±3 d window, and do not replicate this site's earlier 1965–2016 test (±7 and ±15 d: no excess).
- **Negative.** January 2018 (M6+ counts at the 7th–12th percentile) and August 2024 (mainshocks 16th, eruption onsets 9th percentile) were quieter than usual. April 2011 and March 2020 were normal. Supermoons and planet groupings show no excess for any outcome or window.

7. Mechanisms and alternative explanations

Tides are the only physical link of measurable size. Solid-earth and ocean tides change stress on faults by about 0.01–0.1 bar. That is far less than the stress released in an earthquake. Studies find a small tidal influence on some shallow thrust faults [6], and a higher share of large events during high tidal stress [7]. Neither study allows a particular

month or fault to be singled out. A global study of M8+ earthquakes found no preferred lunar phase [8]. A supermoon raises the lunar tide by about 20% over an average full Moon. That is comparable to the ordinary perigee spring tides that come every few months.

Planets: Venus, the strongest planetary tide, is about 1/20,000 of the Moon's at its closest. Jupiter is about 1/100,000 or less. A grouping or conjunction seen from Earth does not change those values in any meaningful way. **Comets and meteor showers** have no known route of influence on the solid Earth at these distances and masses.

Alternative explanations for apparent hits: aftershock sequences, which are handled by declustering and the circular-shift null; ongoing eruptions reported as new; reporting lags; and selection. Selection is the largest: choosing which "alignment" to look at after a disaster has happened. With about 13 M7+ mainshocks a year, any 30-day window has roughly a two-in-three chance of containing one somewhere.

8. Limitations and monitoring plan for 7–30 November 2026

Data cutoff: USGS to 6 Nov 2026; GVP database onsets to 31 Dec 2025. **Observed so far in November** (not a test): M6+ events M6.0 165 km NW of Port-Olry, Vanuatu (2026-11-01 11:47 UTC, aftershock), M6.1 Fiji region (2026-11-05 17:21 UTC, aftershock). Only five historical clusters exist, so the test has little power to detect a weak effect. A small effect cannot be excluded, but no large effect is present. Gardner–Knopoff windows ignore depth and are approximate. Eruption onsets depend on reporting, and escalations and unrest could not be tested.

Monitoring plan, fixed now: on 1 December 2026, rerun `tools/sky/cluster_study.py` with the same definitions on USGS data to 30 November. Report for the 13–27 November window: M6+ counts (all, mainshocks, M7+) against three baselines. (1) Long-term: the same-length windows above (means 5.7 / 4.2 / 0.5). (2) Recent rate: 1 August–6 November 2026 scaled to 15 days (6.5 all, 4.5 mainshocks), to allow for 2026's own activity level. (3) Strong El Niño months, 1973–2025 (ONI ≥ 1.5), scaled to 15 days (3.7 mainshocks). Also report eruption onsets from the GVP Weekly Volcanic Activity Reports, dated by activity start and not by report date, against a mean of 1.2. Any count outside the 95% range of all three baselines will be reported as unusual. It will still not count as evidence of an astronomical cause.

9. Background conditions in 2026

November 2026 is not a quiet background. Three conditions are tested here: a record El Niño (ONI 2.46 in ASO 2026), a northeast Pacific marine heatwave, and an active earthquake year.

- **Earthquake activity in 2026 is elevated but within past range.** To 6 November there were 124 M6+ events (90 mainshocks), against a 1973–2025 mean of 117 (85) for the same part of the year; 20 earlier years had as many or more. M7+ mainshocks number 13 against a mean of 10.3; 10 of 53 years had as many or more. October 2026 was busy, with 21 M6+ events, many of them in the Vanuatu sequence.
- **El Niño:** months with ONI ≥ 1.5 in 1973–2025 (31 months; the 1982–83, 1997–98 and 2015–16 events) averaged 7.6 M6+ mainshocks a month, against 8.3 overall. The difference is not significant ($p = 0.91$, circular shifts of the ONI series). The correlation between monthly ONI and mainshock counts is $r = 0.03$. No ENSO effect on global M6+ seismicity is detected, and no mechanism of the needed size is known. El Niño does matter for coastal hazards: sea level, storm surge, landslides from heavy rain and coastal erosion. Those are tracked on the dashboard, not here.
- **Marine heatwave:** ocean warmth changes stress in the crust by far less than the ordinary tide. There is no known pathway to earthquakes or eruptions, and no data to test one, so it is listed as context only.

These background checks are separate from the astronomical test. They set the expected rate for November, so that a busy El Niño year is not mistaken for an astronomical effect.

Sources

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Data snapshot: 6 November 2026. Independent, non-commercial analysis built from public sources (see About & method). Analysis prepared with Claude (Anthropic). Contact